

3.6 Unit 6 Investigative and Practical Skills in A2 Chemistry

Candidates should carry out experimental and investigative activities in order to develop their practical skills. Experimental and investigative activities should be set in contexts appropriate to, and reflect the demand of, the A2 content. These activities should allow candidates to use their knowledge and understanding of Chemistry in planning, carrying out, analysing and evaluating their work.

The content of Units 4 and 5 provide a basis for different practical topics which may be used for experimental and investigative skills. The experience of dealing with such activities will develop the skills required for the assessment of these skills in the Unit. Examples of suitable experiments that could be considered throughout the course will be provided in the Teachers' Resource Bank.

It is expected that candidates will be able to use and be familiar with 'standard' laboratory equipment which is deemed suitable at A2 level, throughout their experiences of carrying out their practical activities.

The skills developed in the course of their practical activities are elaborated further in the *How Science Works* section of this specification (see Section 3.7).

In the course of their experimental work, candidates should learn to

- demonstrate and describe ethical, safe and skilful practical techniques
- process and select appropriate qualitative and quantitative methods
- make, record and communicate reliable and valid observations
- make measurements with appropriate precision and accuracy
- analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.

The Centre-Assessed Option, Route T

The practical and investigative skills will be centre assessed through two methods

- Investigative Skills Assignment (ISA).
- Practical Skills Assessment (PSA).

The ISA will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test (ISA test) (see Section 3.8).

The PSA will be based around a centre assessment, throughout the A2 course, of the candidate's ability to follow and undertake certain standard practical activities across the three areas of Chemistry: Inorganic, Organic and Physical. These practical activities are listed on the next page.

The Externally Marked Option, Route X

The practical and investigative skills will be externally assessed through

- Practical Skills Verification (PSV) and
- Externally Marked Practical Assignment (EMPA).

The PSV will require candidates to undertake the practical activities identified on the next page, in order to gain experience of a wide range of practical skills. These activities must allow a candidate suitable opportunity to demonstrate safe and skilful practice, as well as producing reliable and valid observations. AQA will require teacher verification, by means of a tick box on the Candidate Record Form, that a candidate is experienced in these skills.

The EMPA, in a similar way to the ISA, will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test (EMPA test) (see Section 3.8).

Practical Skills Assessment

The PSA is designed to credit candidates for the practical work they undertake naturally as part of the course. Further information is provided in Section 3.8, but it is expected that candidates will complete at least two practical activities from each of the three areas of Chemistry listed below.

A2 Inorganic Chemistry

Task	Possible context
Carry out a redox titration	The analysis of iron tablets by titration using acidified potassium manganate(VII)
Investigate the chemistry of transition metal compounds in a series of experiments	The chemistry of copper compounds
Prepare an inorganic complex	The preparation of hexaamminecobalt(III) chloride or the preparation of iron(II) ethanedioate

A2 Physical Chemistry

Task	Possible context
Carry out a kinetic study to determine the order of a reaction	An iodine clock experiment e.g. the reaction of sulfite ions with iodate(V) ions
Determine an equilibrium constant	Determine a value of K_c for the reaction of ethanol with ethanoic acid.
Investigate how pH changes when a weak acid reacts with a strong base or when a strong acid reacts with a weak base.	Determine the pH curve for ethanoic acid reacting with sodium hydroxide

A2 Organic Chemistry

Task	Possible context
Prepare a solid organic compound	The preparation of aspirin
Purify an organic solid	The recrystallisation of impure benzenecarboxylic acid from hot water
Test the purity of an organic solid	Determine the melting point of benzenecarboxylic acid